



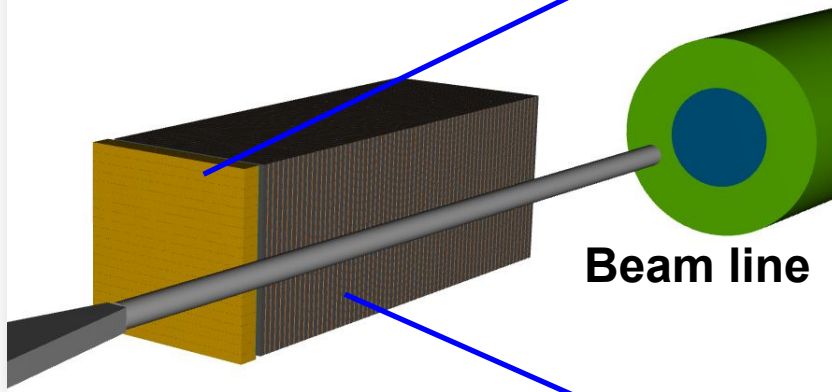
# **ZDC MC Status (Taiwan Group) EIC-Asia @ 20260501**

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**Academia Sinica, Taiwan**

# Official ZDC Design : Crystal ECAL + HCAL

**ZDC = ECAL + HCAL**

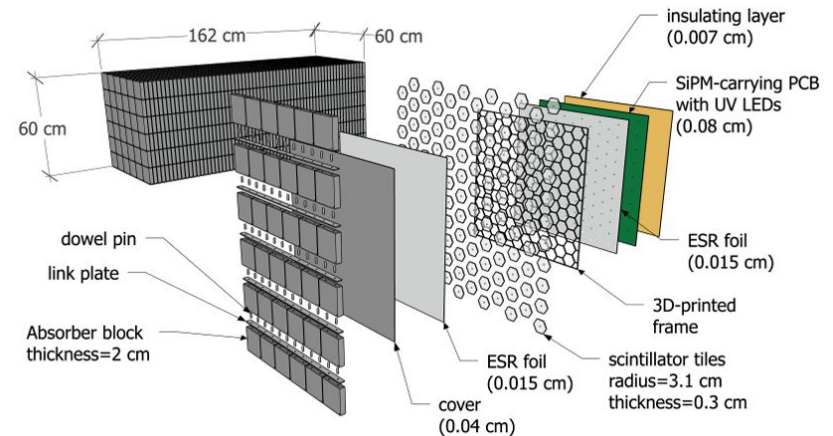
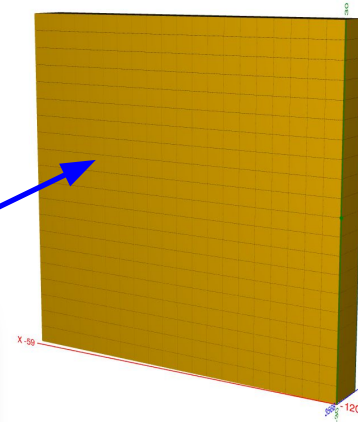


**Beam line**

- Official MC is used :  
ECAL crystal + HCAL SiPM-on-tile

## ECAL

1. **LYSO Crystal**
2. 20\*20 cells
3. 3cm\*3cm\*7cm / cell
4. 60cm\*60cm
5. 7cm ~ 6.5X0 in Z

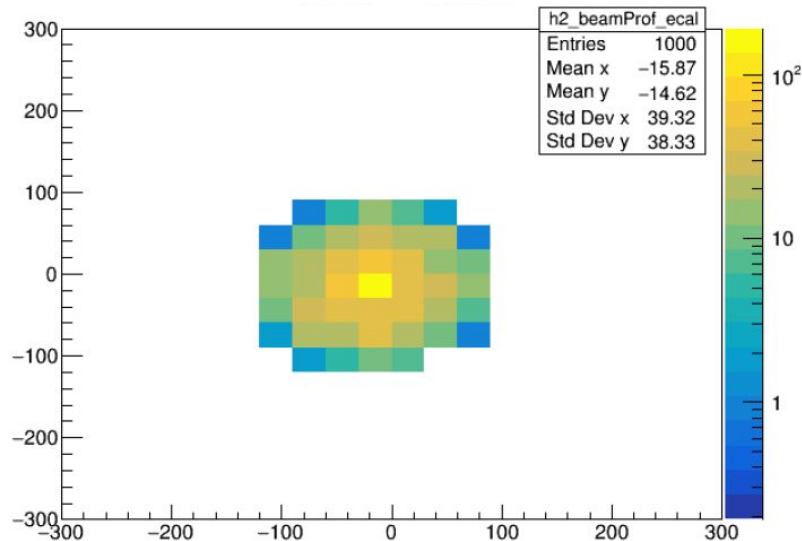


## HCAL : sampling calorimeter

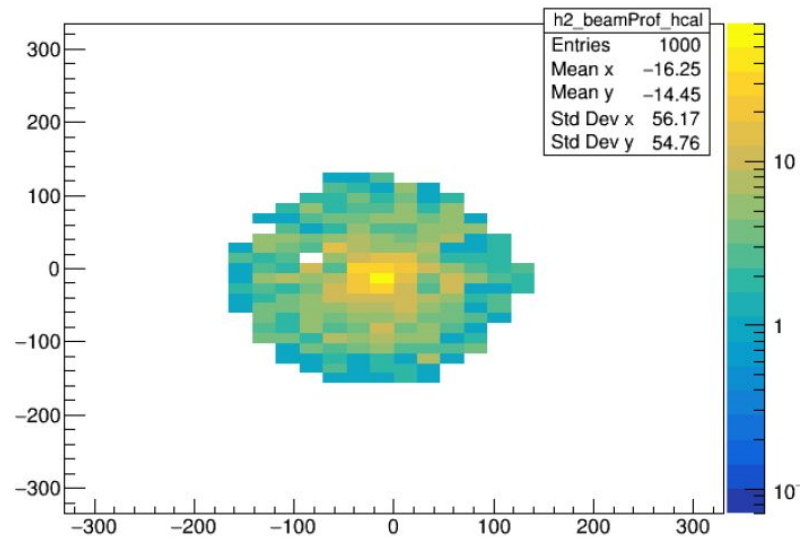
1. **1 layer = steel + scintillator tile + SiPM**
2. 64 layers, 8 slice/layer
3. 65cm in X, 60cm in Y, 163cm in Z

# Relocate Detector and Beam Position

Beam profile on ECAL surface  
(300 GeV neutron)



Beam profile on HCAL surface  
(300 GeV neutron)



- **Only ZDC in MC**
- ZDC is aligned straight along z-dir.
- Gun moves in front of ZDC.
- **Spread beam**
  - **25 degree opening** angle
  - 1 GeV – 40 GeV gamma
  - 20 GeV – 300 GeV neutron



# Energy Reconstruction

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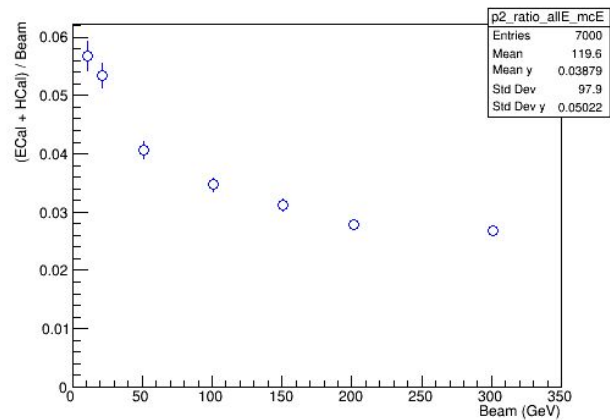
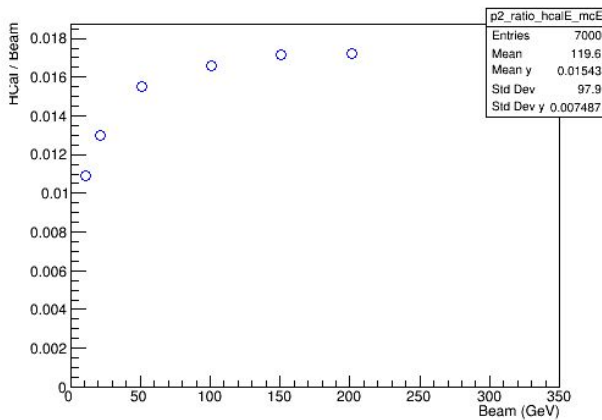
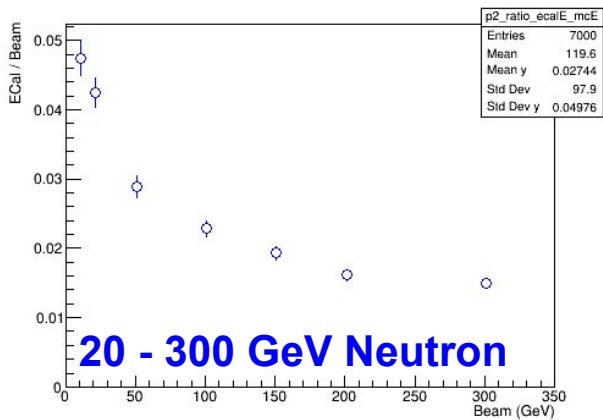
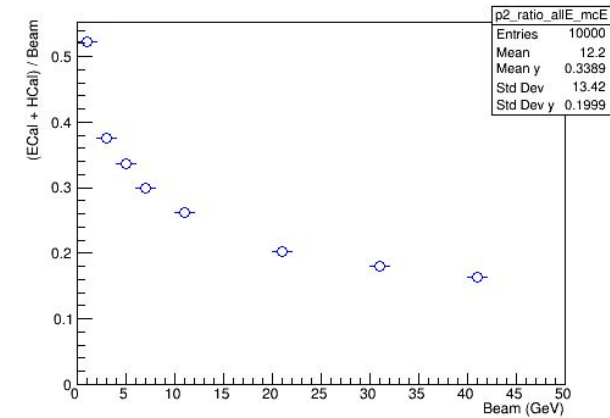
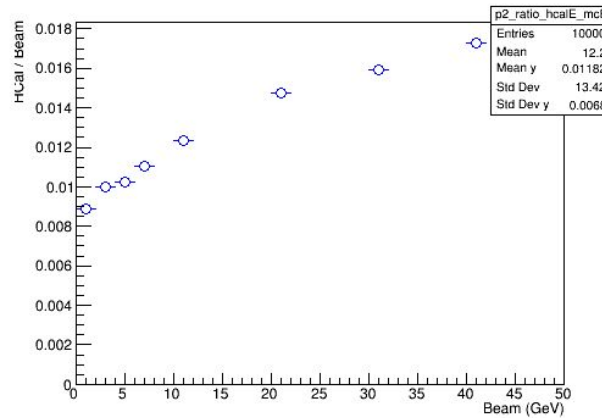
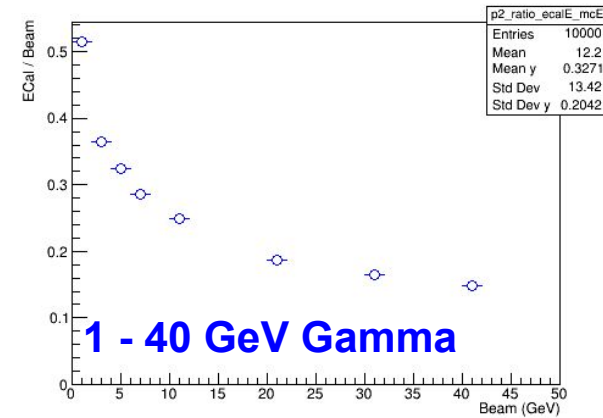
Energy Resolution  
And  
Energy Bias

# Energy Dump

Beam VS ECAL/Beam

Beam VS HCAL/Beam

Beam VS  
(ECAL + HCAL)/Beam



- 1 - 40 GeV gamma beam : Most of energy dumped in ECAL ~ 20% to 60%.
- 20 – 300 GeV neutron beam : Not much energy dumped in scintillator tile, only 3% to 6%.

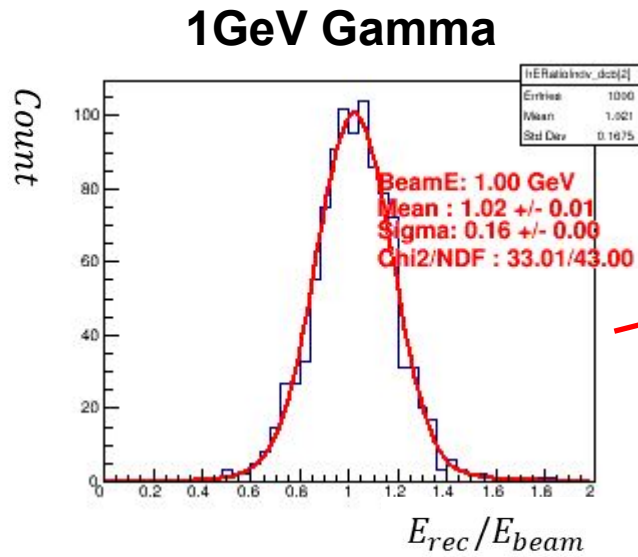
# Different Kinds of Energy Regression

- **Fitting Functions**

- Linear function :  $E_{rec} = p0 + p1 * E_{ECAL} + p2 * E_{HCAL}$
- Linear function w/ ratio :  $E_{rec} = p0 + p1 * E_{ECAL} + p2 * E_{HCAL} + p3 * E_{HCAL}/(E_{ECAL} + E_{HCAL})$
- Quadratic function :  $E_{rec} = p0 + p1 * E_{ECAL}^2 + p2 * E_{HCAL}^2$

- **Weighting (Events enter fitting with a certain weight)**

- Equal weighting : *weighting* = 1 (all the events are equal)
- Square-root of beam energy : *weighting* =  $\sqrt{E_{beam}}$  (higher energy beam has more weight)



## Linear energy regression

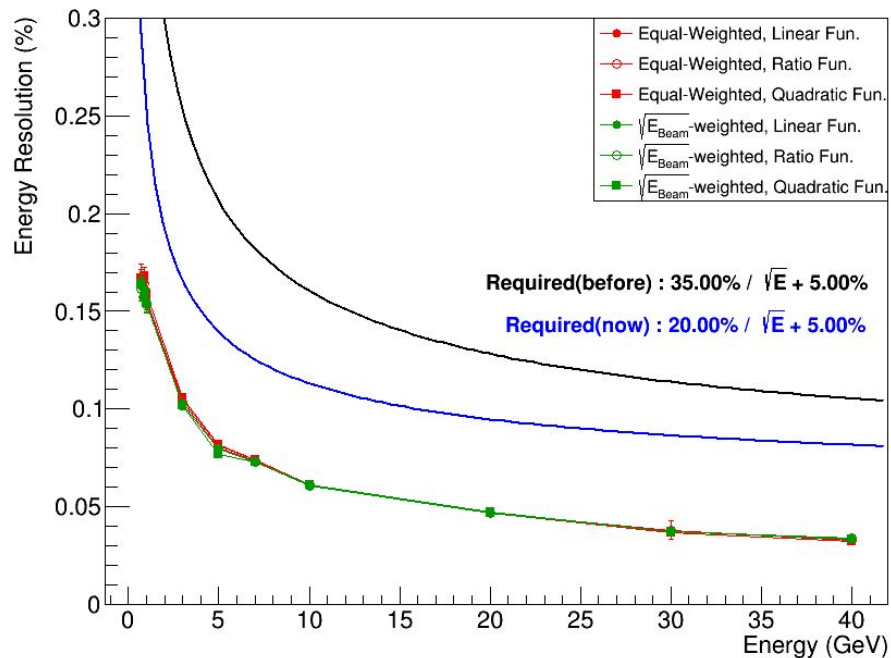
$$E_{rec} = p0 + p1 * E_{ECAL} + p2 * E_{HCAL}$$

Fit  $E_{rec}$  with double - side crystal ball function

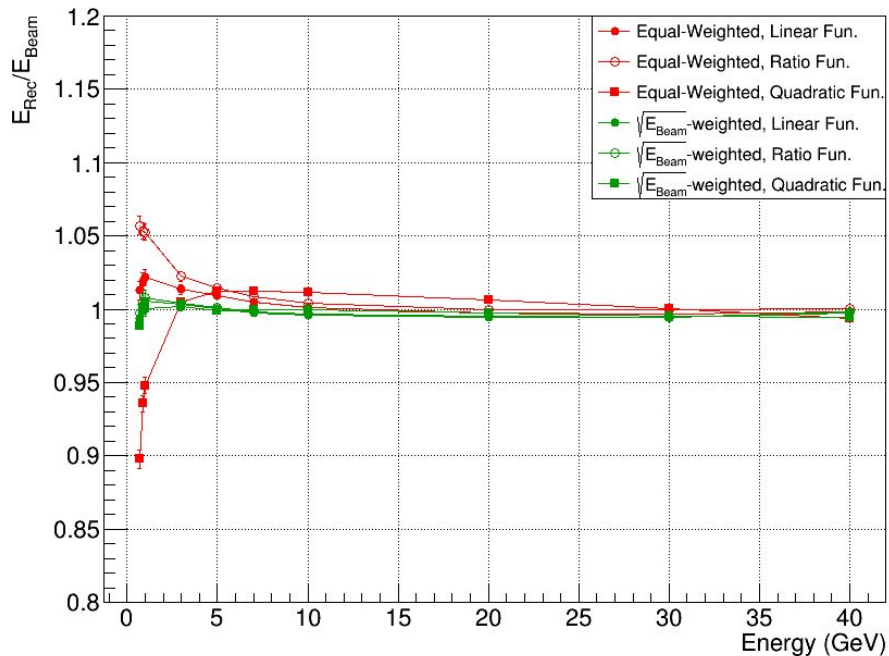
- **Energy resolution** =  $\frac{\sigma}{E_{beam}}$  (%)
- **Energy bias** =  $\frac{\mu}{E_{beam}}$

# Gamma Beam Energy Regression

Energy Resolution



Energy Bias



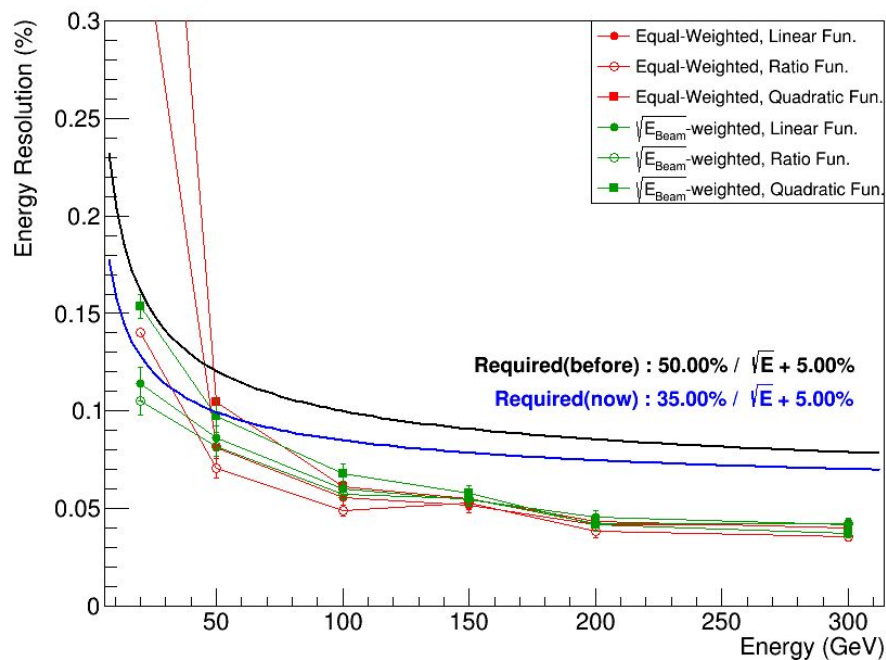
## 1–40 GeV Gamma Beam

- **Method : Sqrt(Ebeam) weighting** performs best. insensitive to fitting function.
- **Resolution:** Performance requirements fully satisfied.
- **Bias:** Negligible energy bias observed.
- **Nice energy reconstruction for gamma.**

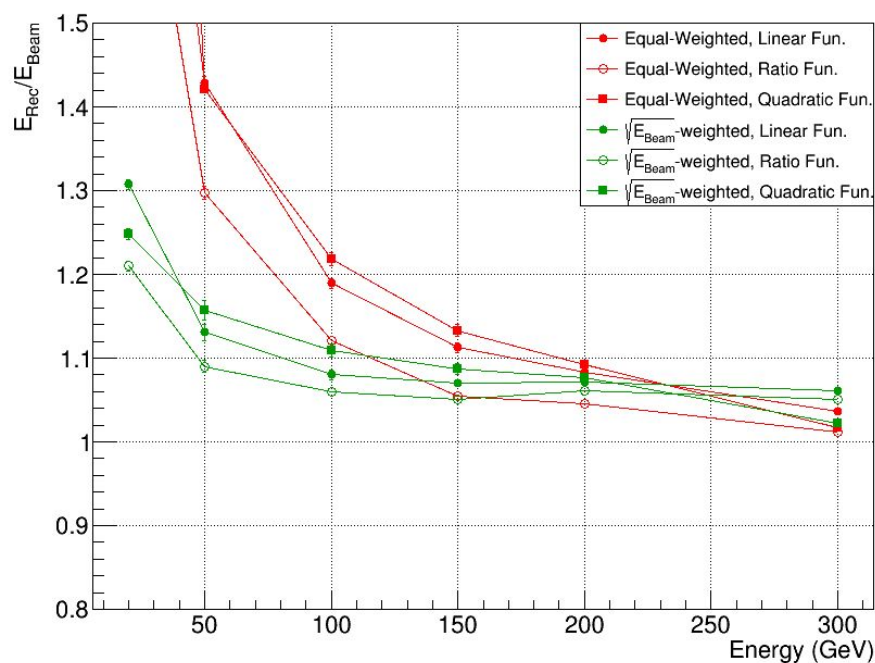


# Neutron Beam Energy Regression

Energy Resolution



Energy Bias



- **Optimal Method: Sqrt(Ebeam) weighting** combined with the **Ratio Method** yielded the best results.
- **Energy Resolution: neutrons < 50 GeV does not meet the requirements.**
- **Energy Bias: A bias of approximately 2% is observed for neutron > 50GeV. Large bias for low energy region (neutron < 50GeV).**
- **The small fraction of neutron energy deposited in the scintillator tile (3%–6%) limits regression precision for beam energies under 50 GeV?**





# Track Reconstruction

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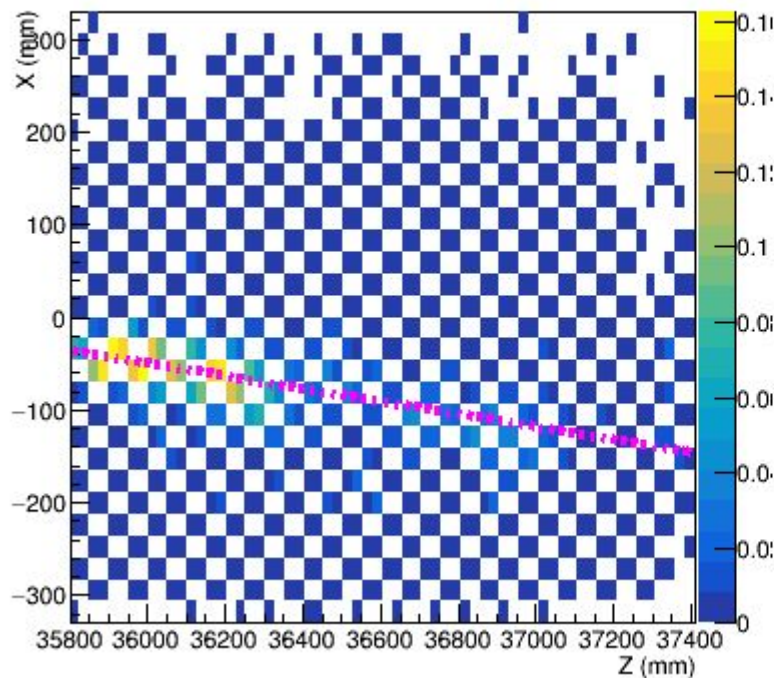
Position Resolution  
And  
Position Bias

# Track Reconstruction Algorithm

## (1/2)

- Only HCAL is used in track reconstruction.
- **Method**
  - Three dimensional space => **XZ** and **YZ** planes.
  - Fit **averaged hit position in layers with polynomial1** to find track.
  - **Residual** can be estimated by compare beam extrapolation and reconstructed track.

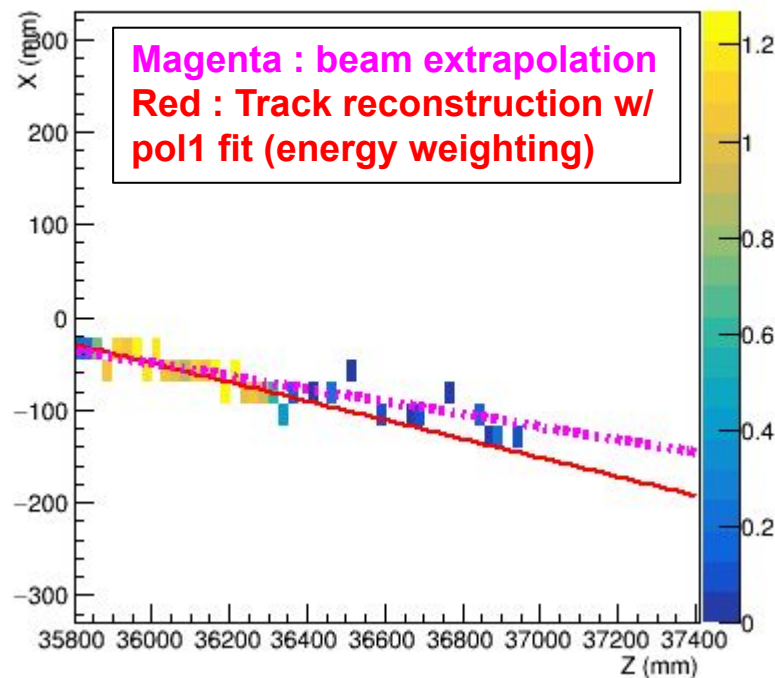
300 GeV neutron beam in HCAL  
Hit position in XZ plane



Fit average  
hit position  
layer by layer

$$\bar{x} = \frac{\sum E_{hit} x_{hit}}{\sum E_{hit}}$$

300 GeV neutron beam in HCAL  
Average Hit position in XZ plane



# Track Reconstruction Algorithm

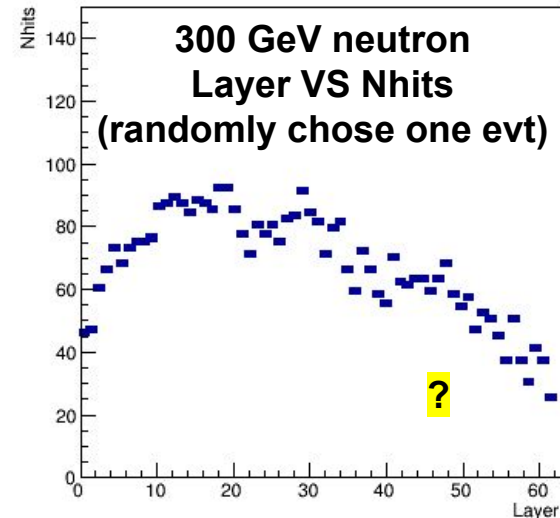
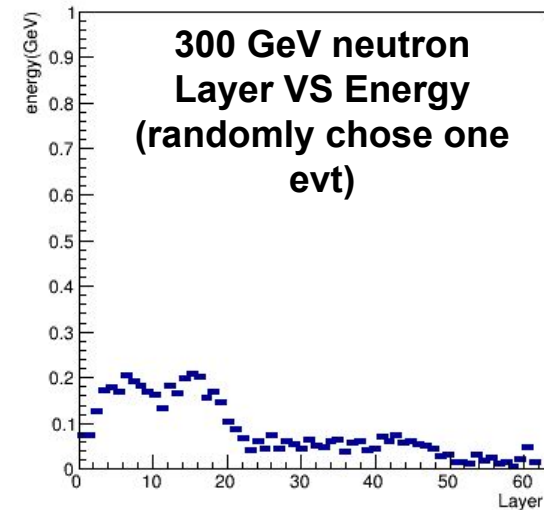
## (2/2)

- Fitting Function : polynomial 1
- Weighting of each layer in fitting
  - ① Linear energy :  $E_{\text{layer}}$
  - ② Log energy :  $\log(E_{\text{layer}})$
  - ③ Hit count :  $\frac{N_{\text{layer}}}{N_{\text{sum}}}$
  - ④ Energy-and-hit :  $\frac{E_{\text{layer}}}{E_{\text{sum}}} + \frac{N_{\text{layer}}}{N_{\text{sum}}}$

$E_{\text{layer}}$  and  $E_{\text{sum}}$  are energy of layer and energy of all layers, respectively.

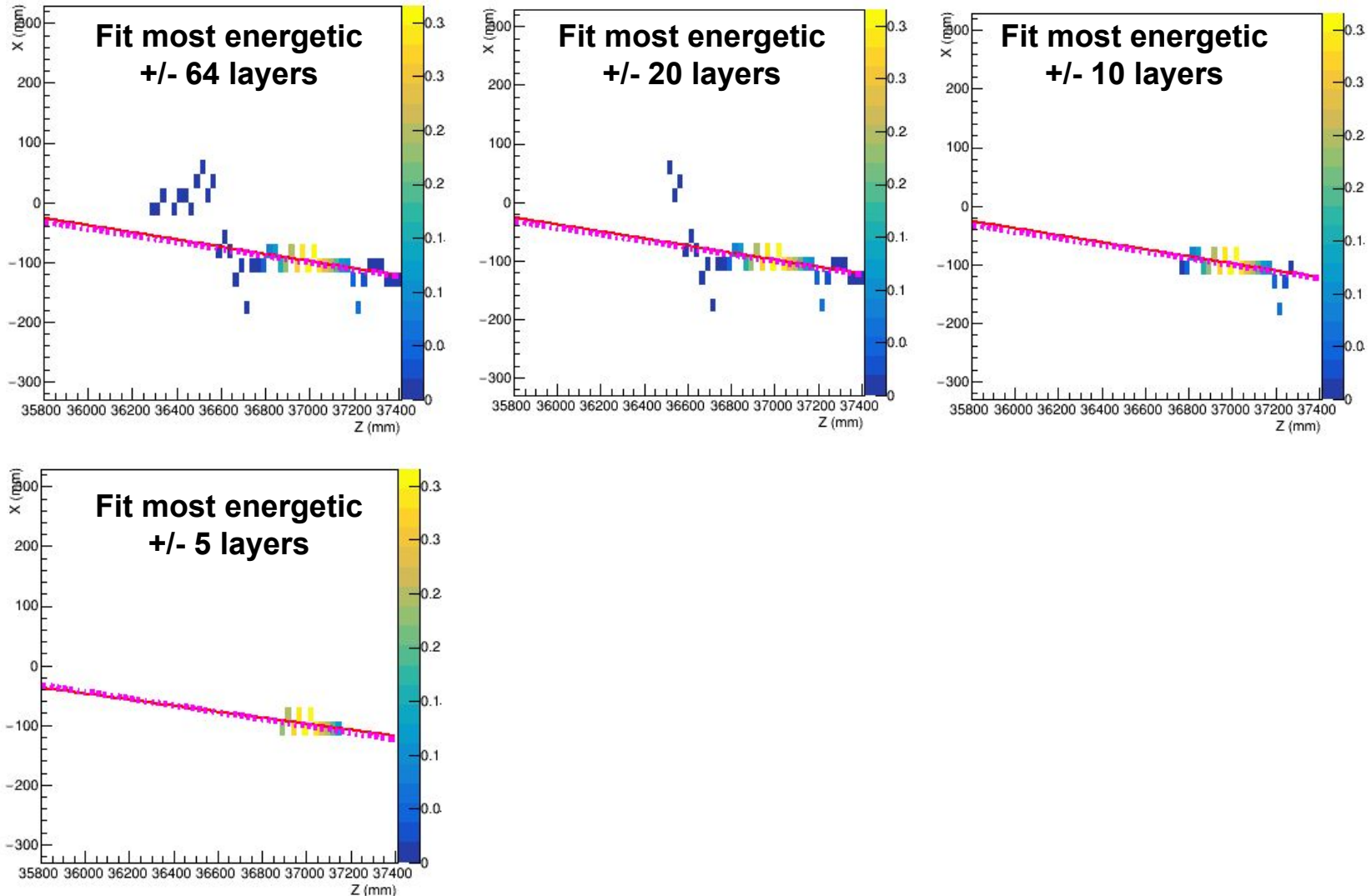
$N_{\text{layer}}$  and  $N_{\text{sum}}$  are num. hits of layer and num. hits of all layers, respectively.

- Cut low energy layers  
Most energetic layer +/- N layers



# Test Different Num. of Fitting Layer (1)

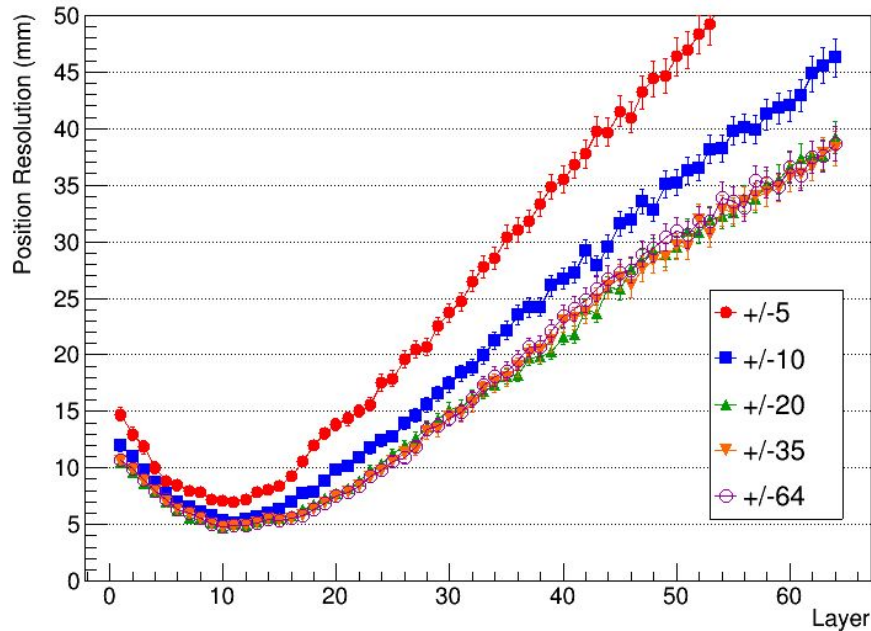
One event from 300 GeV Neutron / Linear Energy Weighting



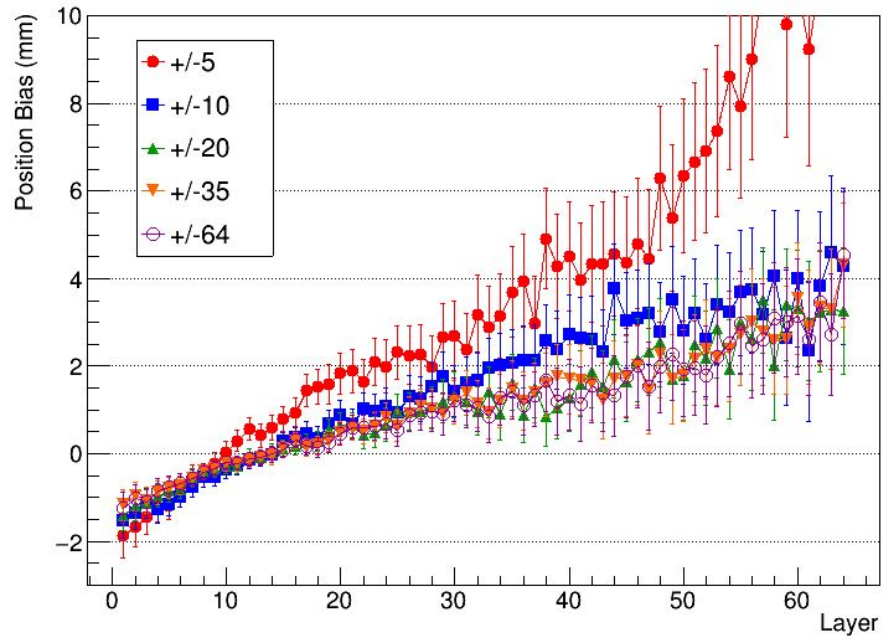
# Test Different Num. of Fitting Layer (2)

300 GeV Neutron / Linear Energy Weighting / XZ plane

Position Resolution



Position Bias

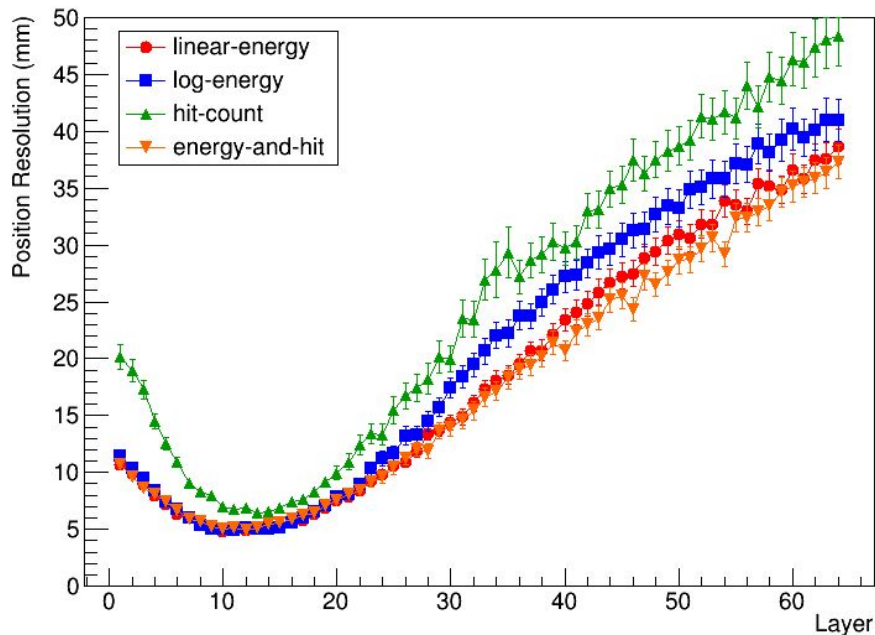


- All results utilize **linear weighting** with a **first-order polynomial (pol1)** fit.
- **Reducing the number of layers in the fit does not improve position residual or bias.** The **weighting** already assigns lower importance lower energy layers, effectively performing the same function as a layer cut.
- **Position resolution and bias degrade as the distance from the shower maximum (the most energetic layer) increases.**

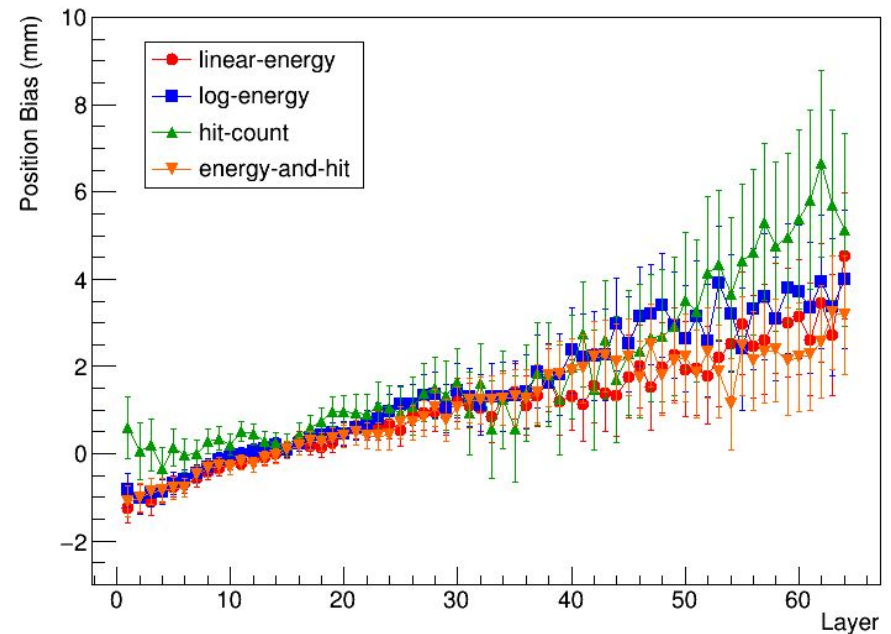
# Test Different Weighting Factor

300 GeV Neutron / Fit 64 layers / XZ plane

Position Resolution



Position Bias



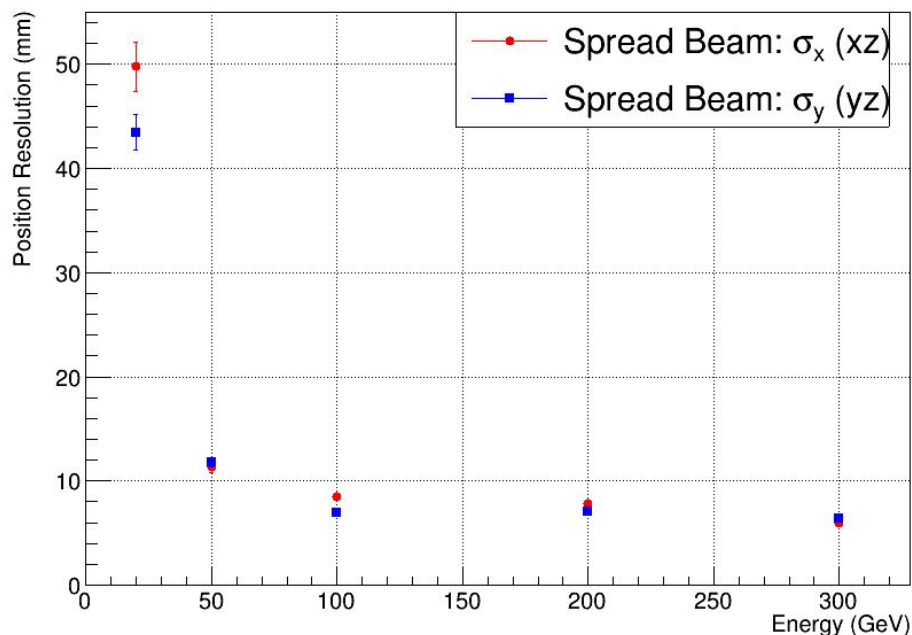
- Except for the hit-count weighting approach, all other weighting methods have consistent results.
- Based on its overall performance and stability, the **hit-and-energy weighting method** was chosen for the final track reconstruction.



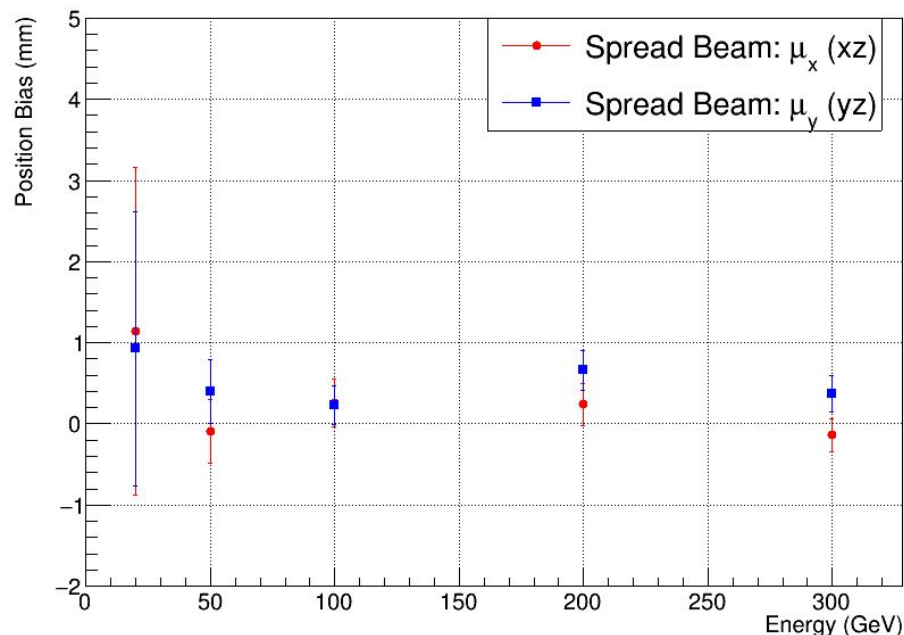
# Neutron Beam Track Reconstruction

## Fit 64 layers / hit-and-energy weighting

### Position Resolution @ Weighted Z position



### Position Bias @ Weighted-Z position



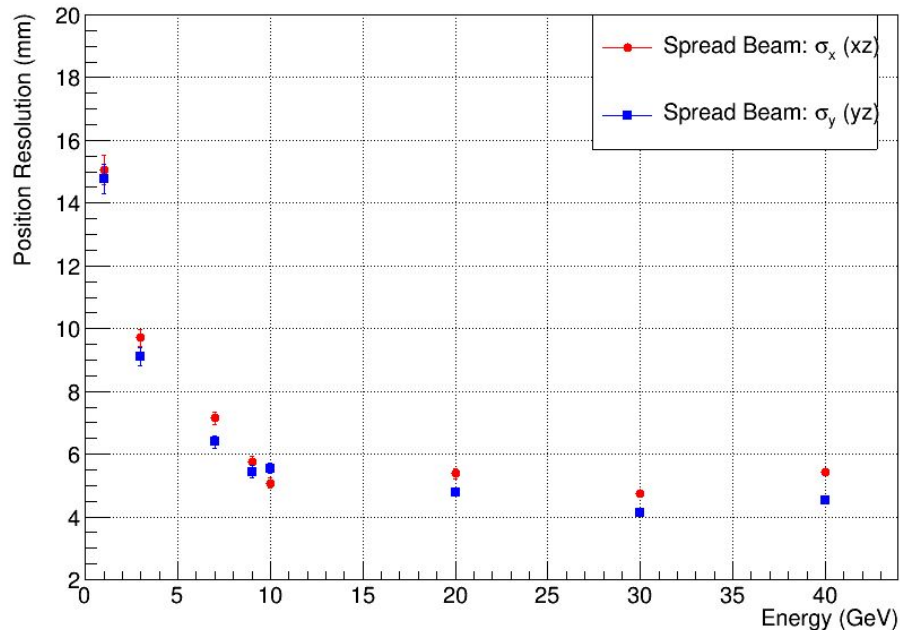
- The results here are shown at **weighted-z position** is close to the most energetic layer.
- **Position Resolution:** Achieved a peak resolution of **6 mm** for high-energy neutrons.
- **Reconstruction Bias:** Minimal bias observed, approximately **0.4 mm**.



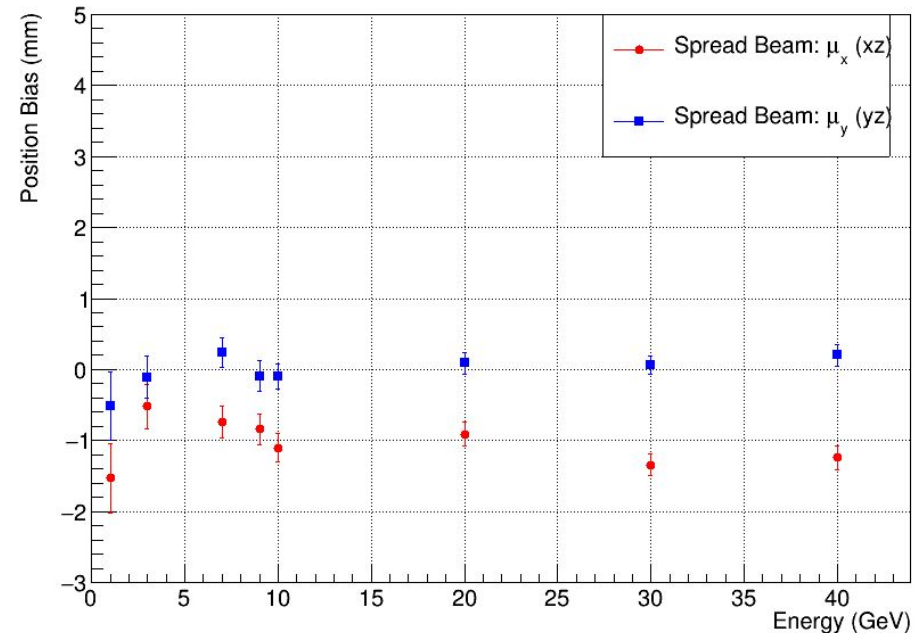
# Gamma Beam Track Reconstruction

## Fit 64 layers / hit-and-energy weighting

### Position Resolution @ Weighted Z position



### Position Bias @ Weighted-Z position



- The results here are shown at weighted z position of all hits which is close to the most energetic layer.
- **Position Resolution:** Achieved a peak resolution of **5 mm** for high-energy gamma.
- **Reconstruction Bias:** Large bias observed for xz plane, approximately **1 mm (reason still unknown)**.

# Kaon/Pion Structure Study

DIS process

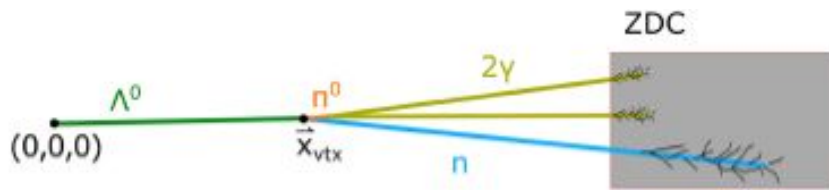
$$|p\rangle = \underbrace{Z^{1/2}|uud\rangle}_{\text{probe proton}} + \underbrace{a|n\pi^+\rangle + b|p\pi^0\rangle + c|\Delta^{++}\pi^-\rangle}_{\text{probe pion}} + \underbrace{d|\Lambda K^+\rangle + e|\Sigma^0 K^+\rangle + f|\Sigma^+ K^0\rangle + \dots}_{\text{probe kaon}}$$

Pion-exchange Sullivan process for pion structure

$$e + p \rightarrow e' + X + \mathbf{n}$$

Kaon-exchange Sullivan process for kaon structure

$$e + p \rightarrow e' + X + \Lambda \quad \mathbf{\Lambda} \rightarrow \mathbf{n} + \pi^0 \rightarrow \mathbf{n} + 2\gamma$$



- It is important to check the position resolution at vertex point, also scan the lambda decay across different Z-positions.
- ZDC sits 35m away from the collision point. From the current results, to reconstruct gamma or neutron track with only HCAL will not be enough for the pion/kaon structure study. WSi ECAL is proposed to help.

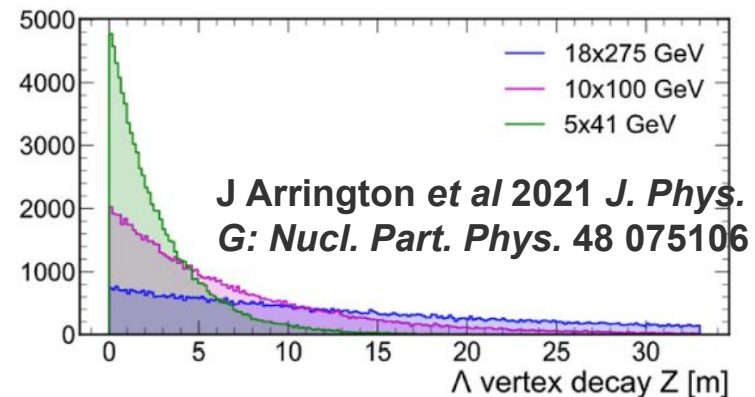


Figure 13. The  $\Lambda$ -decay spectrum along the beamline for different beam energies.

# Summary and To Do



- **MC simulation condition :**

The simulation utilizes a **ZDC (Crystal ECAL + SiPM-on-tile HCAL)** aligned with the z-axis. The setup uses **spread beams** (25 degree opening angle) to evaluate performance for **gammas (1–40 GeV) and neutrons (20–300 GeV)**.

- **Energy reconstruction**

**Nice energy reconstruction for gamma (requirement satisfied).** However, for neutron the small fraction of energy deposited in the ZDC (3%–6%) limits regression precision for beam energies under 50 GeV. **Only neutron > 50GeV satisfied the requirement.**

- **Track reconstruction**

**Position resolution and bias degrade as the distance from the shower maximum (the most energetic layer).** Lambda decay point could be up to 35m away from ZDC. It is important to check the position/vertex resolution across different Z-positions. From the current results, to reconstruct gamma or neutron track with only HCAL will not be enough for the pion/kaon structure study. WSi ECAL is proposed to help.

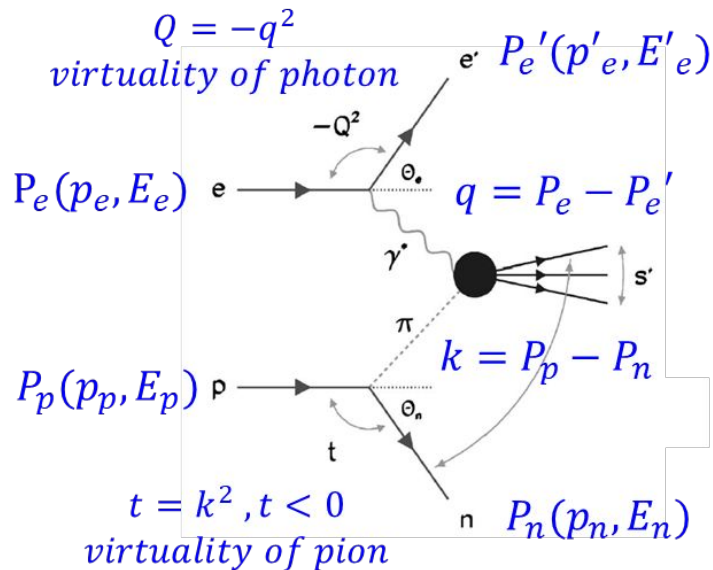
- **To do :**

- **Update HCAL geometry (hexagonal -> rectangular scintillator tile)**
- Evaluate performance with **crystal ECAL + HCAL SiPM-on-tile with neutron/gamma/lambda gun across varying Z-positions.**
- Evaluate performance with **WSi ECAL + HCAL SiPM-on-tile with neutron/gamma/lambda gun across varying Z-positions.**
- Explore **GNN** to and check energy/track reconstruction.



# Back up

# Sullivan Process for Pion Structure Study



## • Pion kinematics

- ①  $t = k^2 = (1 - x_L)m_p^2 - p_T^2/x_L - [(1 - x_L)/x_L]m_n^2$ ,  
pion virtuality,  $t < 0$
- ②  $x_\pi = \frac{Q^2}{(2k \cdot q)} \approx \frac{x}{(1 - x_L)}$ , momentum fraction of parton in pion

## • Neutron observables

- ①  $x_L \approx E_n / E_p$ : Neutron longitudinal momentum
- ②  $p_T \approx x_L E_p \theta_n$ : Neutron transverse momentum

The essential measured quantities are  $x_L$  and  $p_T$  of very forward angle neutron  $\rightarrow$  zero-degree calorimeter.

J.D. Sullivan, "One-pion exchange and deep-inelastic electron-nucleon scattering", Phys. Rev. D 5, 1732 (1972).

The nucleon is not a rigid 3-quark system; it is dressed by a cloud of mesons. Therefore, the proton's wavefunction is expanded not just as a bare  $|uud\rangle$  state, but as a superposition that includes pion-baryon and kaon-hyperon components. The pion and kaon structures can be explored through the Sullivan process from DIS data.

$$|p\rangle = Z^{1/2}|uud\rangle + a|n\pi^+\rangle + b|p\pi^0\rangle + c|\Delta^{++}\pi^-\rangle + d|\Lambda K^+\rangle + e|\Sigma^0 K^+\rangle + f|\Sigma^+ K^0\rangle + \dots$$

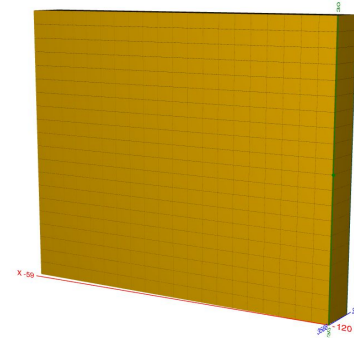
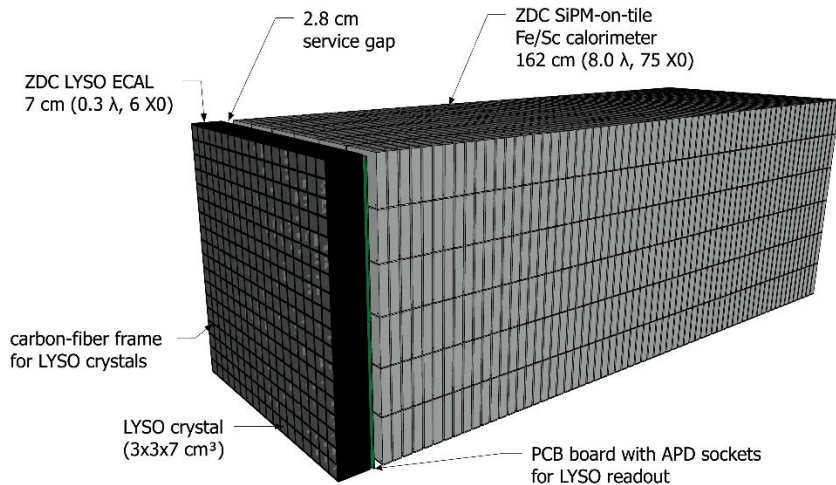
probe proton

probe pion

probe kaon

# Zero Degree Calorimeter (ZDC)

## Current Design



**ECAL**  
**LYSO + SiPM**  
**detect gamma**  
 60cm\*60cm  
 20\*20 cells  
 3cm\*3cm\*7cm / cell  
 7cm ~ **6X0** in Z

**HCAL**  
**sampling calorimeter**  
**detect neutron**

- Pion-exchange Sullivan process for pion structure  

$$e + p \rightarrow e' + X + n$$
- Kaon-exchange Sullivan process for kaon structure  

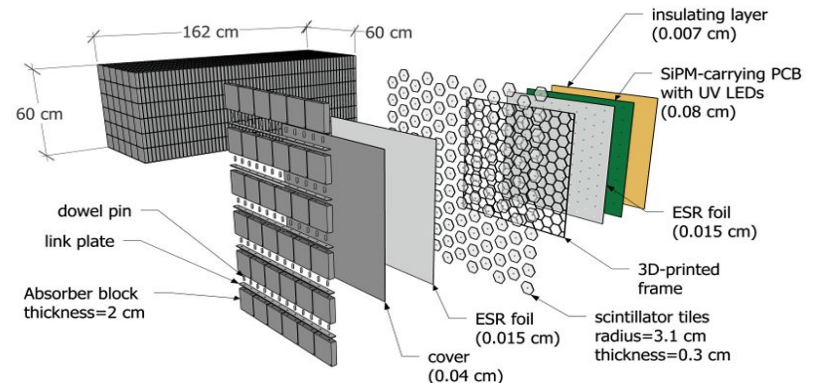
$$e + p \rightarrow e' + X + \Lambda \quad \Lambda \rightarrow n + \pi^0 \rightarrow n + 2\gamma$$
- Spectator-neutron tagging for nuclear physics  

$$e + d \rightarrow e' + X + n$$

$$e + A \rightarrow e' + X + (A - 1) + n$$
- Background, secondary photons showers  

$$\pi^0 \rightarrow \gamma\gamma$$

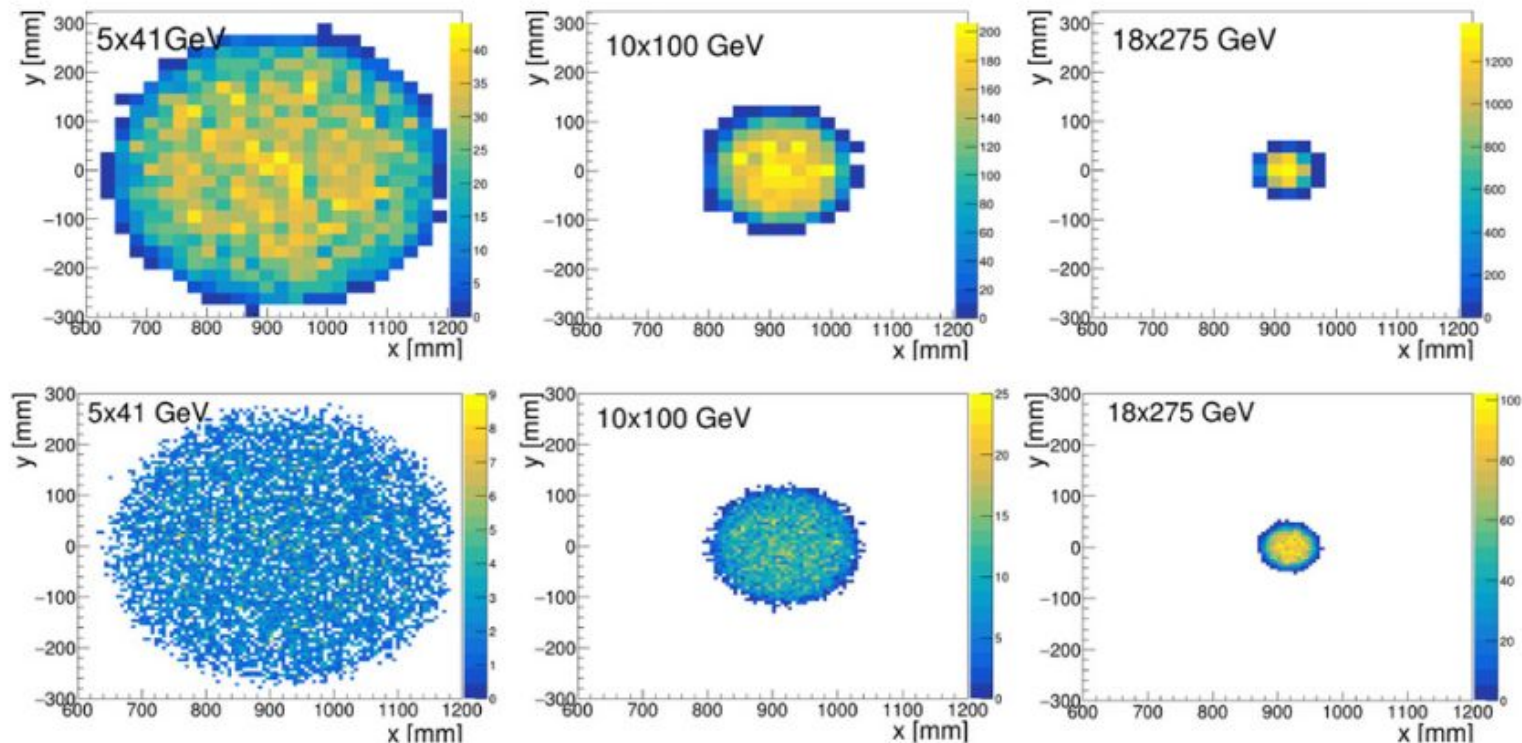
ZDC aims to measure gamma up to 40 GeV and neutron up to 300 GeV. Therefore, it is composed by ECAL and HCAL.



64 layers, 8 slice/layer  
**1 layer = steel + scintillator tile + SiPM**  
 65cm in X, 60cm in Y, 163cm in Z



# ZDC Acceptance



**Figure 12.** Acceptance plot for neutrons in the  $60 \times 60 \text{ cm}^2$  ZDC, with a low spatial resolution of 3 cm (upper panels) and with a high spatial resolution of 0.6 cm (lower panels), for different energy settings, from left to right, of  $5 \times 41$ ,  $10 \times 100$ , and  $18 \times 275$ . The acceptance plot for  $5 \times 100$  would be similar to that shown for  $10 \times 100$ . The lower proton (ion) energies set the requirement for the size of the ZDC, whereas the higher proton (ion) energies drive the spatial resolution requirement.



# Energy Resolution VS Energy

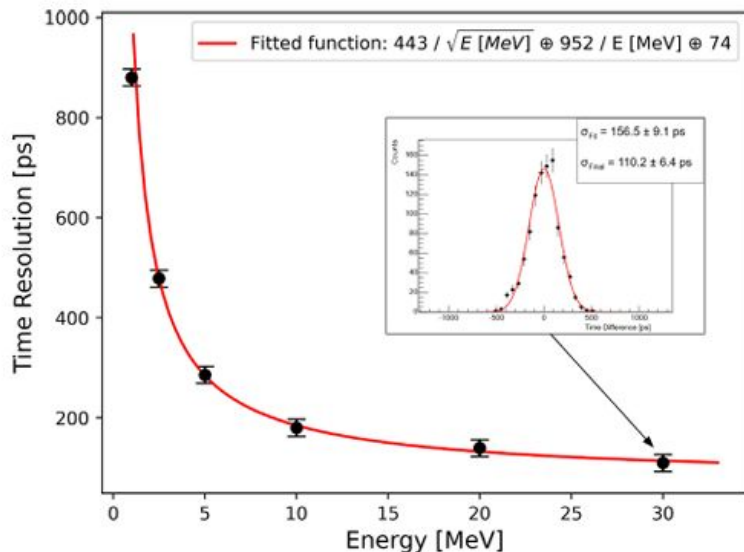


Fig. 13. Dependence of time resolution of a LYSO crystal in the array on the energy deposition in the LYSO crystal for energy deposition between 1–30 MeV. A 110 ps time resolution is measured for 30 MeV energy deposits after the resolution of the reference time is removed via quadrature from  $\sigma_{Fit}$ .

Energy resolution measurements were performed using all three radioactive sources, which provided nine different energies in total.

The resulting energy resolution of a LYSO crystal as a function of  $\gamma$ -ray energy is shown in Fig. 5. An electromagnetic calorimeter energy resolution is typically represented with the following functional dependence:

$$\frac{\sigma_{E_\mu}}{E_\mu} (\%) = \frac{a}{\sqrt{E}} \oplus \frac{b}{E} \oplus c, \quad (2)$$

where  $a$ ,  $b$ ,  $c$  are constants and  $E_\mu$  is the incident particle energy in MeV. Here,  $a/\sqrt{E}$  is a statistical term used to express the contribution from Poisson processes, such as photostatistics, to energy resolution. The  $b/E$  term parameterizes noise contributions to energy resolution from electronics and PMTs, and the constant  $c$  parameterizes contributions from shower leakage, crystal non-uniformity, and intra-crystal miscalibrations to energy resolution. In single crystal testing, the photosensor used to read out the crystal is operated at a high voltage where noise is minimized, thereby resulting in  $b \rightarrow 0$  in the fit to Eq. (2). The constant term  $c$  is also assumed to be dominated by crystal non-uniformity for single crystal tests. We find  $a = (3.84 \pm 0.19) \sqrt{\text{MeV}}$  and  $c = (0.64 \pm 0.57)$ . Because the PIONEER experiment operates at a higher energy range than radioactive sources ( $\mathcal{O}(10) - \mathcal{O}(100)$  MeV), the stochastic term will be greatly suppressed in the PIONEER energy regime.